

Scalability & Future Plan

Date	13 July 2026
Team ID	N/A
Project Name	FinRelief AI – AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the FinRelief AI solution can be scaled to handle larger borrower bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current pilot state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address
1	Single SQLite database	Limits concurrent write performance as the borrower base grows	Medium
2	No caching layer for AI responses	Repeated Gemini API calls increase latency and API cost	Medium
3	Single-server deployment	No load balancing or horizontal scaling under higher traffic	High

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a small pilot group of borrowers on a single instance	Migrate to PostgreSQL with connection pooling and deploy behind a load balancer
2	Data Storage	SQLite file-based storage	Move to a managed PostgreSQL/MySQL instance with automated backups
3	Performance	Synchronous Gemini API calls block requests during letter generation	Introduce an async task queue (e.g., Celery with Redis) for negotiation letter generation
4	Security	Basic JWT authentication	Add refresh tokens, rate limiting, and role-based access control (borrower vs admin)

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment with CI/CD pipeline	Q3 2026	Public accessibility for pilot borrowers

Phase 2	Multi-lender negotiation letter templates	Q4 2026	Improved personalization and higher settlement success rate
Phase 3	Mobile application (Android/iOS)	Q1 2027	Wider borrower reach and easier access
Phase 4	Predictive analytics for early debt-stress alerts	Q2 2027	Proactive borrower support before loans become overdue